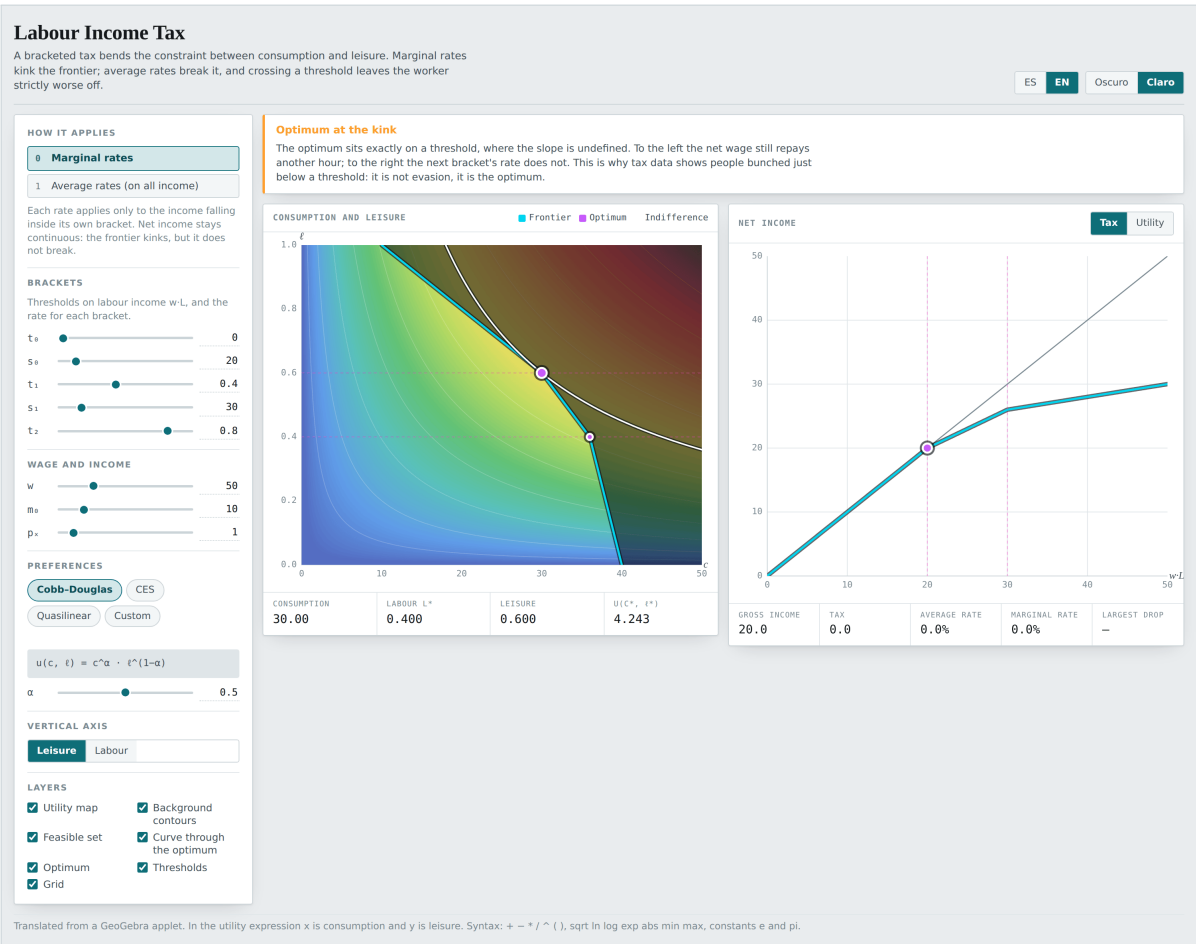


# Labour Income Tax

A bracketed tax bends the constraint between consumption and leisure. Marginal rates kink the frontier; average rates break it, and crossing a threshold leaves the worker strictly worse off.



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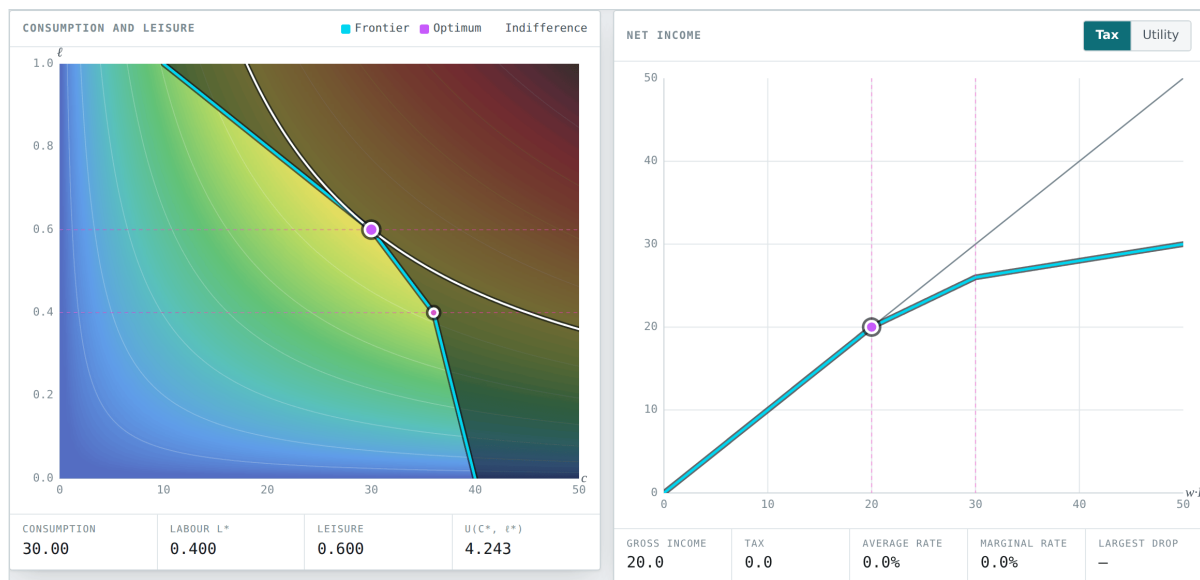
## 1 What this is

A worker splits a time endowment between working and not working. Devoting a fraction  $L$  to work earns  $wL$ , pays a tax on that income, and spends what is left on consumption:

$$c(L) = \frac{m_0 + wL - T(wL)}{p_x}, \quad \ell = 1 - L.$$

The endowment is normalised to 1, so  $L$  is the fraction of time worked and  $\ell$  is leisure. The only thing to choose is  $L$ .

All the interest is in  $T(\cdot)$ , the bracketed tax, and in a design decision that real tax systems make and that classes almost never discuss: whether each bracket's rate applies *only to the income falling inside that bracket* or *to all income*. The two schemes sound nearly identical and produce completely different budget frontiers.



**Figure 1.** Marginal rates. Each bracket's rate taxes only the income inside it. Net income stays continuous, so the frontier *kinks* but does not break.

### The point

Under marginal rates the frontier has a **kink**: the slope changes abruptly but the line does not break. Under average rates it has a **notch**: earning one more unit leaves the worker with less. A kink attracts optima; a notch pushes them down and pins them just below the threshold.

### 1.1 Who it is for

The consumption-leisure choice, or a public-economics class on tax design. It assumes indifference curves, a budget constraint and the idea of an optimum. Without changing anything it also serves to explain why tax data shows people bunched just below a threshold.

## 2 Opening it

There is nothing to install. The whole application is one `index.html` file: it downloads no libraries, calls no server, and stores nothing outside your browser.

### 2.1 The three ways

**Double-click the file.** The quickest route. Find `labor-tax/index.html` and open it. The browser shows it straight off the disk, with an address beginning `file://`. It works offline from the first moment.

**From the web.** If someone has published it, the address ends in `/labor-tax/`. Once the page has loaded you can disconnect: it needs the network for nothing else.

**Served from a folder.** For a class, putting the folder on any static server is enough. With Python installed:

```
python3 -m http.server 8000
```

then `http://localhost:8000/labor-tax/` in the browser.

#### Note

Saving the page with `Ctrl + S` (or `Cmd + S` on a Mac) leaves a copy of your own that still works offline and that can travel on a memory stick or go out by email.

### 2.2 Language and theme

Two pairs of buttons sit at the top right.

| Control      | What it does                                                                                                                                                     |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ES / EN      | Switches the whole interface between Spanish and English on the spot. Nothing is recomputed, so you can switch mid-explanation without losing the state you had. |
| Dark / Light | Switches the theme. Dark is the original design; light exists for projectors and for printing.                                                                   |

Both choices are remembered in the browser, so next time it opens the way you left it.

### 2.3 Which browser

Any reasonably recent one: Chrome, Edge, Firefox or Safari, on a computer, tablet or phone. On a narrow screen the panels stack one above the other instead of sitting side by side. No account, no permissions, no extensions.

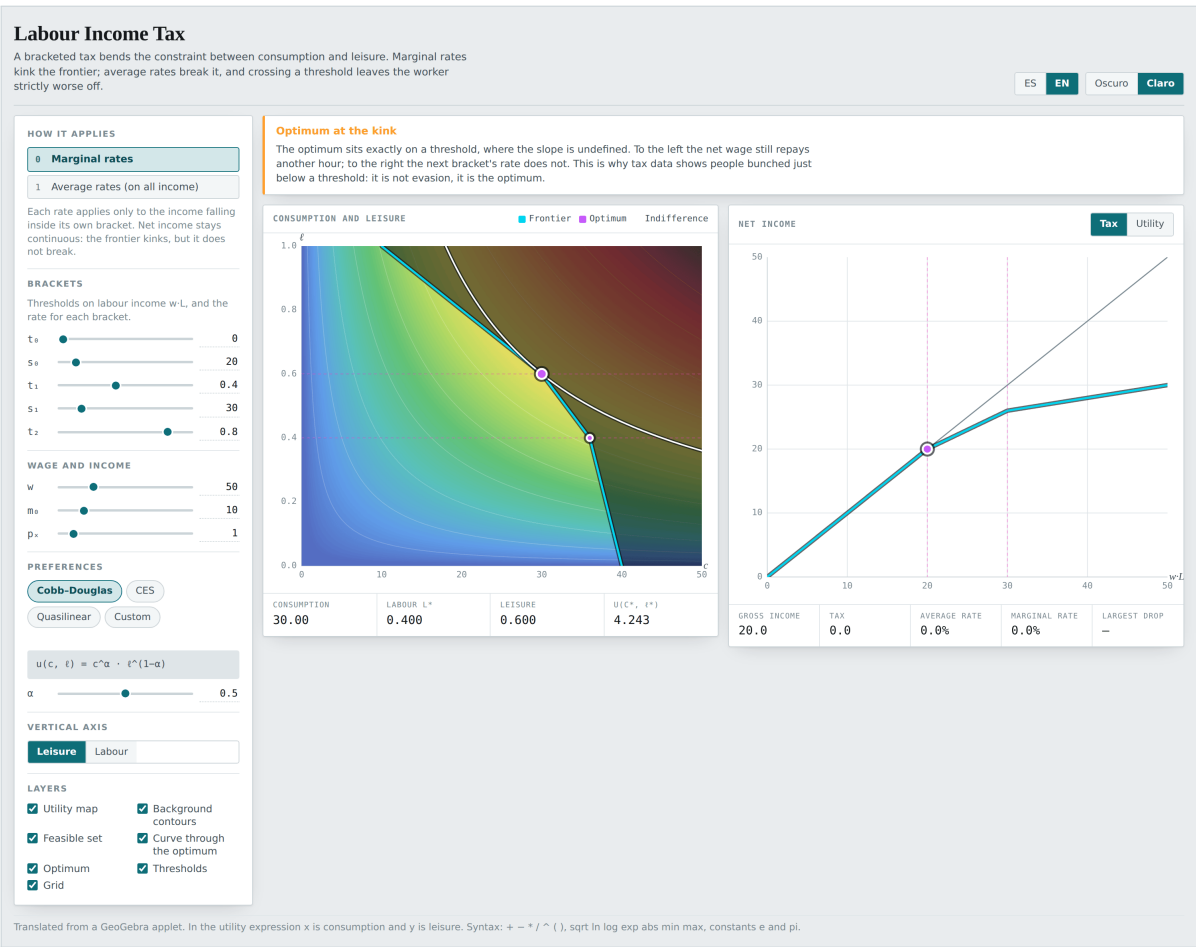
## 3 The screen

**How it applies.** The two schemes, marginal rates and average rates. This is the control that matters; everything else is context.

**Brackets.** Three rates  $t_0, t_1, t_2$  and two thresholds  $s_0, s_1$ , all on labour income  $wL$ .

**Wage and income.** The wage  $w$ , non-labour income  $m_0$ , and the price of consumption  $p_x$ .

**Preferences.** Four families and a box to write your own.



**Vertical axis.** Whether the plane's axis measures leisure or labour.

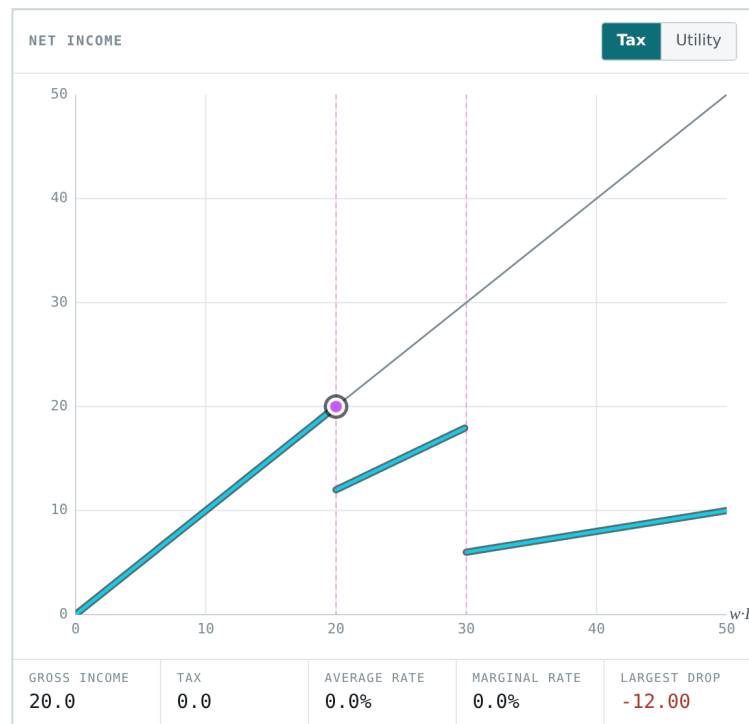
**Layers.** What gets drawn.

The verdict band at the top names which kind of optimum came out: a tangency, a kink, the edge of a drop, not working, or working the whole endowment.

### 3.1 The two panels

**Consumption and leisure.** The utility map under a topographic ramp, everything unaffordable veiled over, the frontier with its kinks or notches, the thresholds, the indifference curve through the optimum, and the optimum.

**Net income.** Net against gross income, with the 45° line for reference. Preferences play no part here, which is the point: the notch is a property of the schedule, not of the worker. The switch changes it to **Utility**, which draws  $u(c(L), 1 - L)$  along the frontier.



**Figure 3.** The net-income panel under average rates. At every threshold net income falls vertically: at the opening values, crossing the first threshold takes it from 20 to 12.

### 3.2 The dark theme

## 4 The model

### 4.1 The two schemes

Let  $t_0, t_1, t_2$  be the rates and  $s_0 \leq s_1$  the thresholds, all on labour income  $Y = wL$ .

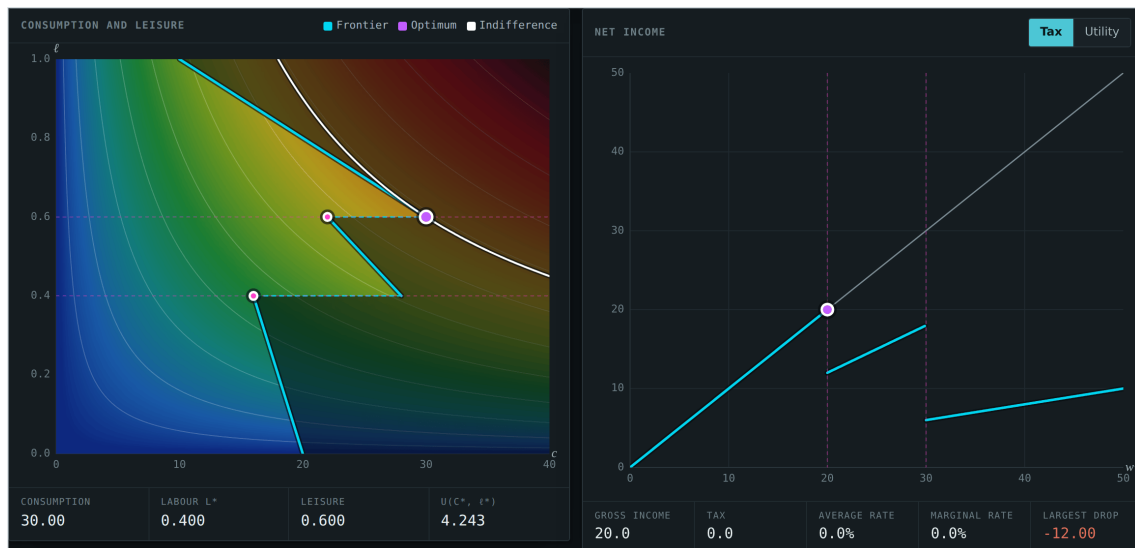


Figure 4. Both panels in the dark theme, under average rates.

### Marginal rates

Each rate taxes only the part of income inside its own bracket:

$$T(Y) = t_0 \min(Y, s_0) + t_1 \max(0, \min(Y, s_1) - s_0) + t_2 \max(0, Y - s_1).$$

$T$  is continuous and increasing, and net income  $Y - T(Y)$  is continuous and concave. The budget frontier is a polyline: at each threshold it changes slope but does not break.

### Average rates

The bracket you land in sets the rate on *all* income:

$$T(Y) = Y \cdot t_{i(Y)}, \quad i(Y) = \begin{cases} 0 & Y < s_0 \\ 1 & s_0 \leq Y < s_1 \\ 2 & Y \geq s_1 \end{cases}$$

Here  $T$  jumps at every threshold and net income jumps down. At the opening values ( $t_1 = 0.4$ ), crossing  $s_0 = 20$  takes net income from 20 to 12: the worker who earns one unit more keeps eight less.

## 4.2 Kink versus notch

The two names are easy to confuse and the difference is everything.

| Name  | What it is                                                                                                                                   |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Kink  | The function is continuous; its derivative jumps. The frontier changes slope. An optimum can land exactly there, and usually does.           |
| Notch | The function itself is discontinuous. The frontier breaks and there is a stretch of points no $L$ reaches. The optimum is pinned just below. |

**Note**

Both generate bunching in the data, but for different reasons: at a kink because the marginal rate changes; at a notch because crossing it simply makes the taxpayer worse off.

**4.3 How the optimum is found**

Not from a first-order condition. On these frontiers the best point is very often at a kink, where the slope does not exist and no tangency holds; and under average rates it sits at the right-hand end of a piece with a strictly worse point immediately to its right.

What happens instead:

1. The frontier is split into pieces, one per bracket, with their own endpoints.
2. Each piece is scanned densely and its best sample refined by golden section.
3. *Every vertex is tested explicitly*, from both sides, each on its own bracket's formula.
4. Whichever wins is the optimum, and where it came from is what classifies it.

**Watch out**

Each piece carries its own bracket rather than looking it up from the income. Looking it up makes  $c(L)$  single-valued at a threshold, which silently loses the left-hand limit — the highest point of the frontier — and left it falling outside the computed view.

**5 The controls****5.1 Brackets, wage and income**

| Slider          | What it is                                                                                                             |
|-----------------|------------------------------------------------------------------------------------------------------------------------|
| $t_0, t_1, t_2$ | The three rates, 0 to 0.95.                                                                                            |
| $s_0, s_1$      | The two thresholds, on labour income. $s_1$ is held at or above $s_0$ : the other way round would invert the schedule. |
| $w$             | The wage. It sets where the thresholds land on the labour axis,<br>$L = s/w$ .                                         |
| $m_0$           | Non-labour income. Shifts the whole frontier upward.                                                                   |
| $p_x$           | Price of consumption. Bounded away from zero.                                                                          |

**5.2 Preferences**

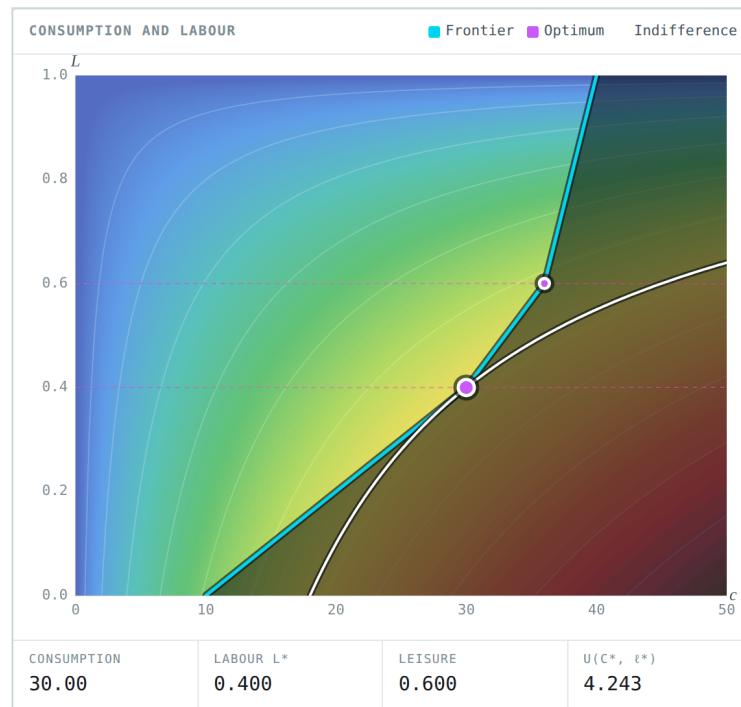
| Family              | Form                                            |
|---------------------|-------------------------------------------------|
| <b>Cobb–Douglas</b> | $x^\alpha y^{1-\alpha}$                         |
| <b>CES</b>          | $(\alpha x^\rho + (1 - \alpha)y^\rho)^{1/\rho}$ |
| <b>Quasilinear</b>  | $x + 4\alpha \ln y$                             |
| <b>Custom</b>       | Whatever you type                               |

In the expression  $x$  is consumption and  $y$  is leisure. With  $\rho$  strongly negative the CES approaches perfect complements and corner optima appear.



### 5.3 Vertical axis and layers

The **Vertical axis** switch moves between **Leisure** and **Labour**. Utility is always read on leisure, so flipping the axis genuinely re-reads preferences rather than mirroring a picture.



**Figure 5.** The plane with the vertical axis set to labour.

| Layer                     | What it draws                                        |
|---------------------------|------------------------------------------------------|
| Utility map               | The field $u$ under a colour ramp.                   |
| Background contours       | Level curves of utility, independent of the optimum. |
| Feasible set              | Veils what cannot be afforded.                       |
| Curve through the optimum | The indifference curve passing through it.           |
| Optimum                   | The chosen point.                                    |
| Thresholds                | The verticals at $L = s_0/w$ and $L = s_1/w$ .       |
| Grid                      | The background rule.                                 |

## 6 The verdict

| Verdict                       | What it means                                                                                              |
|-------------------------------|------------------------------------------------------------------------------------------------------------|
| Interior optimum              | Tangency: the marginal rate of substitution equals the bracket's net wage. The worker is inside a bracket. |
| Optimum at the kink           | It lands exactly on a threshold. To the left another hour still repays; to the right it does not.          |
| Optimum pinned below the drop | Under average rates. The next hour would cross the threshold and levy the higher rate on all income.       |
| Not working is best           | Corner at $L = 0$ : living on non-labour income.                                                           |
| Working the whole endowment   | Corner at $L = 1$ .                                                                                        |

**Optimum at the kink**

The optimum sits exactly on a threshold, where the slope is undefined. To the left the net wage still repays another hour; to the right the next bracket's rate does not. This is why tax data shows people bunched just below a threshold: it is not evasion, it is the optimum.

**Figure 6.** The verdict at the opening values under marginal rates: the optimum lands on the first threshold.

## 7 Classroom activities

### Activity 1 • The same tax, two frontiers

**Settings:** Opening values. Switch between **Marginal rates** and **Average rates** while watching the left panel.

**What you should see:** Under marginal rates the frontier bends at the thresholds. Under average rates it *breaks*: two stretches of line that do not touch, and a whole band of consumption levels no amount of work reaches.

The rates, thresholds, wage and preferences are identical in both. All that changed was the phrase “on the income in the bracket” to “on all income”.

### Activity 2 • Bunching just below

**Settings:** Opening values, **Marginal rates**.

**What you should see:** The **Labour  $L^*$**  readout gives 0.400, which at  $w = 50$  is exactly gross income 20: the first threshold. The verdict says “Optimum at the kink”.

Move  $t_1$  up and down. The optimum stays pinned to the threshold across a wide range of rates, because at a kink there is not one slope to match but a whole interval of them. That is what produces bunching in the data, and it is not evasion: it is the optimum.

### Activity 3 • A notch that costs welfare

**Settings:** **Quasilinear** preferences. Compare **Marginal rates** with **Average rates**.

**What you should see:** Under marginal:  $L^* = 0.800$ , consumption 38.00, gross income 40.0, tax 12.0, utility 34.781. The average rate is 30% and the marginal rate 80%.

Under average rates the same worker retreats to  $L^* = 0.400$ , consumption 30.00, tax 0.0, utility 28.978. It collects *less* tax and the worker is *worse off*. Ask who that design benefits. Nobody: it is pure loss, and it is why almost no tax system uses average rates.

### Activity 4 • Measuring the drop

**Settings:** **Average rates**, right panel on **Tax**.

**What you should see:** The **Largest drop** readout gives  $-12.00$ . Find it in the picture: it is at the second threshold,  $s_1 = 30$ , where the rate goes from 0.4 to 0.8 on all income. Just below, net income is  $30 - 12 = 18$ ; just above,  $30 - 24 = 6$ .

The first threshold drops too, from 20 to 12, but by less. Lower  $t_2$  until the first becomes the largest and watch the readout change.

### Activity 5 • Utility breaks into pieces

**Settings:** **Average rates**, right panel on **Utility**.

**What you should see:** The curve  $u(c(L), 1 - L)$  appears split into disconnected arcs, one per bracket. Its highest point is at the right-hand end of one of them, not at an interior maximum. It is the same information as the left panel, one dimension down. Ask where a student who differentiated and set to zero would look, and why they would not find it.

#### Activity 6 · Not working

**Settings:** CES preferences,  $\rho = -6$ .

**What you should see:** The optimum goes to  $L^* = 0.000$ : consumption 10.00, which is exactly  $m_0/p_x$ . The verdict says “Not working is best”.

With  $\rho$  strongly negative the CES approaches perfect complements and leisure becomes nearly indispensable. Raise  $w$  and see how far it has to go before the first hour is worth it.

#### Activity 7 · The axis is not the preference

**Settings:** Opening values. Move **Vertical axis** from **Leisure** to **Labour**.

**What you should see:** The utility map is not the mirror image of the previous one. Utility is always read on leisure, so flipping the axis genuinely changes what is being drawn, not just how it is being looked at.

A useful check against the habit of reading the two axes as interchangeable.

#### Activity 8 · Designing a tax without notches

**Settings:** Average rates. The goal: a frontier that does not break.

**What you should see:** There is only one way: make all three rates equal. Any difference between  $t_0$ ,  $t_1$  and  $t_2$  produces a notch at the corresponding threshold.

Under marginal rates, by contrast, any increasing combination gives a continuous frontier. There is the whole design argument, and it fits in a sentence.

## 8 Expression syntax

In the utility box,  $x$  is consumption and  $y$  is leisure.

| Element                        | What is allowed                                                         |
|--------------------------------|-------------------------------------------------------------------------|
| <b>Operators</b>               | $+$ $-$ $*$ $/$ $^$ $( )$                                               |
| <b>Functions</b>               | sqrt ln log log10 exp abs sin cos tan min max pow                       |
| <b>Constants</b>               | e, pi                                                                   |
| <b>Implicit multiplication</b> | 2xy is $2*x*y$                                                          |
| <b>Associativity</b>           | $^$ binds to the right and tighter than unary minus: $-x^2$ is $-(x^2)$ |

Partial derivatives are computed symbolically. With min, max or abs — precisely the kinked cases — it falls back to central differences.

## 9 Changes from the GeoGebra original

**One tax base.** TAX\_LABOR taxed  $wL$ , but the average-rate branch 11 taxed  $wL + I_0$ . The two schemes

were therefore not comparable at exactly the moment you want to compare them. Both tax labour income here.

**Discontinuities are modelled.** The original wrote the schedule as nested `If[...]`, which GeoGebra draws with a segment bridging the gap. Each bracket is a separate piece here, so a notch stays a notch.

**Each piece carries its own bracket.** Looking the bracket up from the income makes  $c(L)$  single-valued at a threshold, which silently loses the left-hand limit — the highest point of the frontier. It was falling outside the computed view and being clipped.

**Preferences and an optimum.** The original drew the budget set only. Without an optimum the kink and the notch are shapes rather than predictions, so a utility function, the indifference curve through the optimum and the bunching verdict are added.

$p_X$  and  $w$  started at 0, dividing by zero throughout. Both are bounded away from it.

$s_1 < s_0$  would invert the schedule; the second threshold is held at or above the first.

**Dead objects.** `s_2` had a slider and a label but never entered `TAX_LABOR`; `a`, `b`, `f`, `ngreso`, `E`, `E_trace`, `Pendiente`, `B_OCIO`, `B_TRABAJO`, `h_TRABAJO` and the Q/P point set were leftovers from the applet this was copied from. Not carried over.

## 10 Troubleshooting

| Symptom                                        | What is happening                                                                                             |
|------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| The plane is blank                             | The utility expression does not parse. The red message under the box says whether it is syntax or a variable. |
| The thresholds are not visible                 | They are off the axis: with a small $w$ , $s/w$ can exceed 1. Raise $w$ or lower the thresholds.              |
| Both schemes give the same answer              | All three rates are equal. With equal rates there is neither kink nor notch and the two schemes coincide.     |
| Largest drop reads “—”                         | You are on marginal rates, where by construction there are no drops.                                          |
| The optimum does not move when I change a rate | It is pinned at a kink. That is the correct behaviour and the subject of activity 2.                          |

## 11 Publishing it for a class

Any static host will do: GitHub Pages, Netlify, a university web directory, even a shared folder served over HTTP. The only thing to upload is the `index.html` file, inside a folder named however you want the address to read.

The repository ships a GitHub Actions workflow (`.github/workflows/pages.yml`) that publishes every applet at once on each push to the default branch. To switch it on, in the repository: *Settings* → *Pages* → *Source: GitHub Actions*. That is the one step the workflow cannot do for itself, because creating a Pages site needs administration rights that a `GITHUB_TOKEN` is never granted.

### Note

To hand the applet to students without reliable internet, send them the `index.html` file directly. It weighs less than a photograph and opens with a double-click.

## 12 Licence and provenance

The application code is MIT. Use it, change it and pass it on, in class or out of it, with attribution.

This applet is a standalone rewrite of an original GeoGebra applet. The rewrite is not a literal translation: where the original had a construction error, a division by zero reachable with its own sliders, or a dead object inherited from whatever applet it was copied from, this version fixes it and says so. The section *Changes from the GeoGebra original* lists those changes one by one.

Everything is hand-written and dependency-free: the expression parser, the symbolic differentiation, the level-curve tracing and the drawing. In particular there is no `eval`, so what a student types into a text box cannot become code, and a strict content security policy does not break the page.